

CPSC 554K

Design Review Report #2

Immersive VR Training

Antony Hodgson

March 27, 2023

Derakhshan, Pegah, pegahd@student.ubc.ca
Liu, Zhe, zheliu92@cs.ubc.ca
Prajapati, Himani, himani27@student.ubc.ca
Shirvani, Merry, mehrshi@cs.ubc.ca

1. Overview

1.1 bWell at Home:

Cognitive rehabilitation through Virtual reality (VR) involves the delivery of cognitive exercises which require brain activation in terms of dual tasks, attention span, memory, etc. Much research has been conducted in this area showing numerous benefits in improving cognitive abilities. However, although multiple digital interventions, including VR interventions, have shown considerable promise for digital cognitive health, these interventions have not yet been widely adopted. One reason for this limited adoption is that most VR applications require manual reconfiguration to support repeated measures, which normally only support limited and specific displays and interaction hardware. (Yip and Man, 2013). This complicated operation barrier leads to the demand for trained professionals when delivering cognitive exercises to potential users. On the contrary, the wide acceptance of VR applications (Optale et al., 2010) becomes evident that cognitive rehabilitation with VR could serve as a promising strategy, with the underlying concept on the premise that repeated practice of tasks targeting deficits can lead to improvements in specific cognitive domains. Currently, it remains unclear whether these improvements could easily translate into real-world functioning, for instance, in-home settings for people with mental health disorders, the most common being depression.

Our stakeholders on this project, Cognitive Health Technologies (CHT) team at the National Research Council Canada, aim to fill this gap by developing bWell, a VR application for people with depression. Members of CHT implemented bWell according to the inputs from domain experts and the findings collected from the design process, including matching clinical intentions with VR exercise and configuring and validating multipurpose exercises for specific use cases (Gagnon Shaietz, 2021). At this stage, bWell was implemented as an application in a clinical setting, with a therapist-facing user interface designed for task administration, dosing, and activity data logging. The CHT team would like to transfer bWell into a patient-facing application majorly used in a home setting. In other words, the next version of bWell is expected to be self-guided for patients without extra training and instructions from clinicians. Through this collaboration with DFP@UBC, the CHT team would like our team to explore the potential of bWell used by patients at home for cognitive rehabilitation and remediation, with two major focuses: usability evaluation and long-term engagement.

1.2 Our Progress:

We kicked off this project by conducting a literature review, through which we gathered information on Cognitive behavioral therapy and using VR at home, explored the usability heuristics of VR applications such as an adapted version of Nilsen's heuristics, and investigated potential ways to maintain long-term engagement through the hooked model. Namely, for the latter, we concluded that the use of email notifications could be a low-cost solution to act as an external motivation trigger.

Our literature review was followed by a cognitive walkthrough based on the 2D version of bWell with a semi-structured interview. In this first phase of our study, we received overall positive feedback from the participants. Further, this test, through thematic analysis, revealed general usability issues related to element placement and user flow that were not specific to a 3D interface. We found major fields of improvement in the onboarding, tutorial, and exercise section of the application, including lack of sufficient or well-placed information, page navigation issues, learnability issues and struggles in recalling tutorial information, and mapping issues.

In this report, we will discuss the study's second phase, which involved expert heuristic evaluation. We will provide a detailed account of the process, along with our findings, in their respective sections. Additionally, we will outline our plan for future steps, which includes our third and final phase of the study - the usability test.

2. Problem Statements / Objectives

Applying configurable exercises that pair standardized assessment with adaptive and gamified variants for therapy, cognitive rehabilitation, and remediation has much potential to improve specific cognitive domains and translate these improvements to real-world functioning through repeated practice. To ensure that bWell is suitable for use at home, our long-term objectives focus on answering the following questions:

1. **Potential usability issues:** Is bWell intuitive and easy to use as a self-administrative tool, especially for people with little to no prior experience with VR? What are some common usability issues users face when using bWell (software usage) or the VR headset (hardware usage)?
2. **Maintaining long-term engagement:** How do we maintain users' motivations to engage with bWell long-term?

At this stage, our short-term objective was to conduct an expert evaluation of the product to highlight some usability issues and explore potential solutions. Moving forward, our short-term objective is to evaluate the user experience of bWell through a usability study.

3. Heuristic Evaluation

As the heuristic evaluation is a frequently-used usability inspection method for identifying usability problems in the user interface (UI) and user experience (UX) design, we decided to apply this evaluation method internally within our team before the official usability test to first generate a fundamental understanding on the usability of bWell. In this section, we will cover the general evaluation design, basic evaluation results, and the summarization of the significant findings.

3.1. Evaluation Plan:

We selected Jakob Nielsen's 10 usability heuristics for our evaluation because they are widely recognized and established as a standard for assessing user interfaces. Additionally, research has demonstrated that these heuristics are applicable not only to traditional UIs but also to video games and virtual reality interfaces (Joyce, 2021), which is particularly relevant for our cognitive game-based VR application, bWell.

To obtain comprehensive insights into bWell's interface design, we devised a plan to conduct individual evaluations with each team member acting as a usability review expert, followed by a group discussion to combine our findings. Given our background knowledge in HCI, all of us four could be considered qualified evaluators. Furthermore, having four evaluators falls within the recommended range for the number of evaluators for Jakob Nielsen's heuristic evaluation. To minimize group confirmation bias, we deliberately avoided lengthy discussions about the interface prior to each person completing conducting an independent evaluation of the bWell prototype.

We prepared two sets of materials for the heuristic evaluation of bWell, namely the "Heuristic Evaluation Task List" and "Heuristic Evaluation Form," which can be found in Appendix A1.1 and A1.2. The Task List outlines the entire process of using bWell at home alone, including starting and exiting the VR application, exploring the onboarding process, learning from tutorials, and practicing with the eight exercises. We intentionally included the steps for VR usage and setup to determine whether users with little or no experience with VR devices can access bWell directly. If the results show that users may have difficulties using VR devices, we would further explore identifying specific preparation, introduction, or training that should be provided for first-time users. The Form is designed for each evaluator to document usability issues related to specific heuristic evaluations. To ensure that each evaluator has the same understanding and interpretation of each heuristic, especially in the context of VR applications, we created an additional form containing the definition, example, and interpretation of each heuristic in VR scenarios.

Limited by the delivery of the VR headset between team members, the whole evaluation continued for around one week from Mar 10th to 17th. According to the self-report, each evaluation lasted for 30 to 45 min.

3.2. Evaluation Results:

After consolidating all individual evaluation reports, we identified a total of 78 usability issues, with severity levels ranging from 1 to 4, listed in Appendix A1.3. (A severity level of 0 indicates no usability issues, while a level of 4 signifies a usability catastrophe that must be addressed before product release.) It's worth noting that some of these 78 issues overlap, as multiple evaluators may have identified the same problem. In addition, we included additional labels to the existing columns in the Evaluation Form for each usability issue, including the evaluator who identified the issue and the relevant phase of the user experience (such as onboarding, tutorial, exercise, and others). Since the tutorial of each exercise in bWell follows a similar format to the exercise with additional explanations and instructions, we labeled the usability issues that happened in both phases as "tutorial", because first-time users of bWell would encounter these issues during the tutorial first.

We consolidated similar usability issues within each heuristic identified by multiple evaluators and arrived at 64 distinct issues that cover all 10 heuristics. Of these, three heuristics merit particular attention, as they contributed to a significant number of issues:

- 16 issues arose from a lack of clear visibility regarding the system status
- 14 issues arose from a mismatch between the system and the real world
- 8 issues arose from a lack of adherence to consistent design standards
- Each of the remaining seven heuristics contributed 3-5 issues each.

Next, we examined which phases of the user experience were most frequently associated with these issues. Our analysis revealed that:

- 20 issues were related to the onboarding phase
- 23 issues were related to the tutorial phase
- 13 issues were related to the exercise phase
- 8 issues were related to other phases, such as VR device setup and controller usage

This result aligns with our expectations, as the tutorial phase typically involves more issues than the exercise phase due to our label strategy, and general VR usage can be challenging for novice users without prior experience with VR.

During our discussion, we realized that certain issues were addressed by multiple heuristics and could be resolved with a single design recommendation. Additionally, some issues were more critical than others and required prioritized attention. Consequently, rather than examining each issue individually within the context of each heuristic, we opted to concentrate on the specific phase of the user experience and the part of the interface where the issues occurred. This approach allowed us to identify several key insights that will inform our next-phase redesign efforts.

3.3. Finding Summarization:

As we discovered that 6 of the issues were linked to the initial use of the VR headset and hand controller, we recommend that designers develop a separate manual outlining the necessary procedures that users must complete before starting the bWell application. This manual could be presented as a help documentation resource that includes visual aids such as demonstration images or video illustrations to facilitate the onboarding process.

Regarding the onboarding phase, we found two main issues that require attention. Firstly, some functions, such as the scheduler and sensor, are not yet implemented but remain interactive to empty pages, causing frustration when users get stuck. To prevent errors, designers should consider disabling incomplete functions. Secondly, the personal settings interface needs improvement to enhance the user

experience. Specifically, settings such as avatar skin color and body type, which only apply to specific exercises, should be moved to the exercise-specific settings page to avoid confusion. Additionally, settings such as hand and eye level are not clearly explained, leading to uncertainty about their meaning and expected input. To address this, we suggest adding help documentation with question mark shortcuts for users to access.

We identified issues with both the application tutorial session and the tutorial mode for each exercise, labeled as "tutorial" issues. The most significant problem with the tutorial session is that it is mixed with other exercises, and users may not recognize it as a general introduction to all interactions used in the application. To address this issue, the tutorial session should be made mandatory even before the personal setting page, and the guidelines provided in this tutorial session should be more intuitive for new users.

While the tutorial mode for each exercise is generally helpful in explaining the exercise, triggering it is not easy. Additionally, the duplicated tutorials in the introduction page, video tutorial, and text tutorial may increase users' cognitive workload, requiring redesign.

Two issues that require higher priority attention in the exercise phase are the progress indicator and quit interaction. Three out of four evaluators pointed out that users feel like the exercise tasks are endless without any feedback on the system's status. This issue can be resolved by adding a progress indicator. Currently, when users want to end the exercise, they have to use the "Quit" operation, which is poorly designed. This operation often causes users to quit the entire application instead of just the exercise. Furthermore, the current interaction, which involves head movement and eye gaze, is inconsistent with other frequently used interactions in bWell.

4. Usability Test

4.1. Study Protocol:

In this step, we aim to conduct a formal study that evaluates the effectiveness of the bWell product as a self-administered tool and assesses its user flow while identifying potential usability issues. To achieve this, we have developed a comprehensive guide and protocol for the study that includes essential information and an introduction about bWell and headset usage, scenarios, tasks, and interview questions. The guide is structured to ensure a clear and organized testing process.

To begin the study, we have broken down the user flow into three steps: onboarding, tutorial, and exercise, with an additional pre-step for using the VR headset. The study will be conducted using a think-aloud observation method, starting with an introduction and pre-test questions that gather demographic and general information. The session will be audio/video recorded with the participant's consent, and their interactions with the interface will be recorded and visible to the researchers during the testing by connecting the headset to an external monitor. The session will take about 60 minutes, and we will end the session with a semi-structured interview; thus, the team has the privilege of asking participants to expand on potential interesting points from the observation or the interview itself. The full document of the protocol can be found in the appendix.

4.2. Pilot Study:

To ensure that our protocol is efficient, clear, and effective in gathering the data we need, we have conducted a pilot study. Additionally, we looked for potential technical issues that might arise during the session. Due to time constraints, we have run the pilot once, and our findings are as follows:

a. Technical and Headset Issues:

- There was some confusion on how to use the glass spacer and the possibility of using the headset with glasses on. We need to accommodate the needs of participants that will need to use their glasses during the study.

- We faced some challenges in connecting the headset through the app and cable.

b. Time:

- The time of the pilot was just below one hour, which matched our expectations, with about 15 minutes for each of the pre-test and post-test interview questions and about 30 minutes for the study.

c. Clarity of Instructions:

- Some of our instructions were not clear enough, resulting in the participant asking for clarification or trying to figure it out on their own. Namely, the function and mapping of each button and more detailed task steps within the tutorial.
- Getting stuck on a screen becomes easily overwhelming for participants, given the immersive experience. We need to account for potential edge cases and be prepared to help the participants.
- Some exercises have unexpected features, such as audio notification of time in the theatre exercise, that were not initially included in our instruction.
- We need to clarify that this is a work-in-progress version of bWell and not the final product; thus, there are some placeholders instead of real content which participants would need to ignore.

d. Expected Data

- We were able to gather the expected data from the observation
- Our questions for long-term engagement were a little confusing and would benefit from rewording

5. Next Steps:

Going forward, we plan to conduct usability testing after obtaining approval from our partners and the course instructor team. To ensure that our usability test yields accurate and valuable data, we will improve the study design based on pilot results, which includes refining instructions, clarifying or accounting for any interface elements that may cause confusion, and altering the steps involved if necessary.

Finally, we will integrate insights gained from each phase of our study - literature review, Figma cognitive walkthrough, expert heuristic evaluation, and usability testing - to provide suggestions for a redesign that will enhance user experience.

6. References

- Gagnon Shaiget, V., Proulx, C., Cabral, A., Choudhury, N., Hewko, M., Kohlenberg, E., ... & Debergue, P. (2021). An immersive and interactive platform for cognitive assessment and rehabilitation (bWell): design and iterative development process. *JMIR Rehabilitation and Assistive Technologies*, 8(4), e26629.
- Hernandez Hernandez, M. E., Michalak, E., Choudhury, N., Hewko, M., Torres, I., Menon, M., Lam, R. W., & Chakrabarty, T. (2022). Co-design of a virtual reality cognitive remediation program for depression (bwell-D) with patient end-users and clinicians: A qualitative study (preprint). *JMIR Serious Games*. <https://doi.org/10.2196/43904>
- Joyce, A. (2021, August 11). *10 usability heuristics applied to virtual reality* Alita Joyce. Nilsen Norman Group Website. Retrieved March 26, 2023, from <https://www.nngroup.com/articles/usability-heuristics-virtual-reality/>
- Larson EB, Feigon M, Gagliardo P, Dvorkin AY. Virtual reality and cognitive rehabilitation: A review of current outcome research. *NeuroRehabilitation* 2014 Jun 26;34(4):759-772. [doi: 10.3233/nre-141078]
- Optale G, Urgesi C, Busato V, Marin S, Piron L, Priftis K, et al. Controlling memory impairment in elderly adults using virtual reality memory training: a randomized controlled pilot study. *Neurorehabil Neural Repair* 2010 May;24(4):348-357. [doi: 10.1177/1545968309353328]
- Yip BC, Man DW. Virtual reality-based prospective memory training program for people with acquired brain injury. *NeuroRehabilitation* 2013 Feb 28;32(1):103-115. [doi: 10.3233/nre-130827]

Appendix

Team Contribution

All members were active in conducting individual heuristic evaluation and generating evaluation results	
Team member	Role
Pegah	● Designed study protocol including introduction for the test, tasks scenario and steps ● Completed individual heuristic evaluation ● Contributed in writing study protocol and pilot study section ● Conducted a pilot study and report the result ● Editing the citation of the report ● Formatting the report and submission
Himani	● Designed study protocol including introduction and interview guide. ● Completed individual heuristic evaluation. ● Contributed to writing background, rationale, and objectives. ● Editing the report with proper citations. ● Formatted the report. ● Organizing and editing the appendices.
Merry	● Designed study protocol including researcher guide, introductions, and pre-test questionnaire ● Designed part of the post-test questionnaire and revised questions ● Completed individual heuristic evaluation ● Designed and made the presentation ● Contributed to writing “Our Progress” section ● Revised “Objectives” section ● Wrote “Next Steps” section ● Contributed in writing “Usability Study” section ● Final formatting of the document and grammar check ● Contributed in References and the Appendix organization
Zhe	● Designed the Heuristic Evaluation Task List ● Designed the Heuristic Evaluation Form ● Completed individual evaluation and generated evaluation results ● Merged individual evaluation results and Analysis the significant usability issues ● Contributed in the “Heuristic Evaluation” part in the presentation ● Formatted the report structure ● Wrote section 2. Heuristic Evaluation in the report ● Contributed in the Appendix content and organization

A1 Heuristic Evaluation Materials

In order to enhance readability, we have opted to provide hyperlinks to the relevant documents instead of incorporating the entire content in an extensive appendix. Please kindly note that all documents are available in the original compressed submission as well.

A1.1 Heuristic Evaluation Task List

[Link:](#)  Heuristic Evaluation Task List

A1.2 Heuristic Evaluation Form

[Link:](#)  Heuristic Evaluation Form

A1.3 Heuristic Evaluation Merged Result

[Link:](#)

https://docs.google.com/spreadsheets/d/e/2PACX-1vTx4XnfVilT40fY9G0PYJWQroV7z4-6qiWMec5a7vbki6VDiQc1kpeQHGSxq_9_f3AGZSNKkXgHbWp-/pubhtml?gid=1854828974&single=true

A2 Usability Test Materials

A2.1 Study Protocol

[Link:](#)  Study Protocol - P#